MIL-C-40039D
20 October 1981
SUPERSEDING
MIL-C-40039C
2 May 1968

MILITARY SPECIFICATION

CLOTH, COATED, NYLON, VINYL COATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for vinyl coated nylon cloth. (see 6.1)

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics.

MILITARY

MIL-C-577 - Cloth, Twill, Nylon, 1.6 and 3.0 ounce.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

☆U.S. GOVERNMENT PRINTING OFFICE: 1981—505-022/8432

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-406 - Plastics; Methods of Testing.
FED-STD-601 - Rubber; Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1487 - Glossary of Cloth Coating Imperfections.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified at the time of submission of a bid, the bidder shall submit to the contracting officer a certified copy of a recent laboratory report covering material which he proposed to deliver. Unless otherwise specified by the contracting officer, the bidder shall certify that the material was manufactured in a plant where the coating will be performed if a contract is awarded. This laboratory report shall contain test data which demonstrates that the finished product which the contractor proposes to deliver has been tested in conformance with and found to comply with the requirements of this specification. Any of the following types of reports shall be satisfactory from the standpoint of this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

(a) An independent or commercial laboratory report.
(b) The prospective contractor's own laboratory report.
(c) A governmental laboratory report from a contract within six months of date of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of this specification. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor for subsequently meeting all requirements of the specification on all deliveries.

3.2 Samples.

3.2.1 Standard sample. The coated cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.3 Materials.

3.3.1 Base cloth. The base cloth shall conform to type I of MIL-C-577.

3.3.2 Coating compound. The coating compound shall be polymerized or copolymerized virgin vinyl chloride resin, plasticized with phosphate or phthalate ester plasticizers exclusively, and pigmented to meet the requirement herein.

3.4 Coated cloth. The coating compound specified in 3.3.2 shall be applied to both sides of the base cloth, and the face side shall be more lightly coated than the back (see 6.5). The face side of the base cloth is the side with the twill line running from left to right. The coated cloth shall conform to the requirements of table I and 3.4.1 thru 3.4.4 when tested as specified in 4.4.3.

TABLE I. Physical requirements

Characteristic	Requirement	
	Minimum	Maximum
Weight, oz/sq. yd	6.0	7.3
Breaking strength, pounds		
Warp	90	-
Filling	80	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements (cont'd)

Characteristic	Requirement Minimum	Maximum
Tearing strength, grams		
Warp	768	-
Filling	576	-
Hydrostatic resistance, p.s.i.		
Initial	40	-
After strength of coating	20	-
After water immersion	20	-
Adhesion of coating, lbs./2-inch width		
Initial	6.0	-
After water immersion	5.0	-
Stiffness, centimeters (warp only)		8.0
Blocking, scale rating		No.3

3.4.1 Resistance to abrasion. No loose fibers of the base cloth shall be exposed in the center 1 inch of the abraded area when tested as specified in 4.4.3.1.

3.4.2 Resistance to leakage. There shall be no leakage when tested as specified in 4.4.3.1. Leakage is defined as the appearance of water at three or more different places within the 4-1/2 inch diameter test area.

* 3.4.3 Resistance to low temperature. The coating shall show no indication of cracking, flaking or separation from the fabric after testing as specified in 4.4.3.1.

* 3.4.4 Resistance to accelerated weathering. The coating shall show no increase in softness, stiffness, tackiness or brittleness when compared with an unexposed specimen, after testing for resistance to accelerated weathering as specified in 4.4.3.1.

3.5 Color. The color of the coated cloth shall be OG-207 and shall match the standard sample for shade and luster (see 6.3).

* 3.5.1 Matching. The color and luster shall match the standard sample (see 6.3) under artificial daylight having a color temperature of 7000 $\pm$ 500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 $\pm$ 100 kelvin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

* 3.6 Width. The minimum overall width after selvage trimming shall be as specified (see 6.2). Selvages shall be trimmed to give a straight uniform edge.

3.7 Length. The coated cloth shall be in rolls as specified in PPP-B-1136, of not more than 175 nor less than 150 yards in length. No roll shall contain more than six pieces and no individual piece shall be less than 6 yards in length. The ends of the pieces shall be overlapped, not joined by a seam.

* 3.8 Odor. The coated cloth shall be free from any objectionable odor. Characteristic odor of coating is not considered objectionable.

3.9 Workmanship. The finished coated cloth shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall examined for the defects specified in 4.4.2.1 and tested for the characteristics specified in table III. The presence of any defect or failure of any test shall be cause for rejection of the first article.

* 4.4. Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. A certificate of compliance shall be furnished with each shipment or lot stating the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

a. The coating compound is a polymerized or copolymerized virgin vinyl chloride resin.
b. The plasticizer used is a phosphate or phthalate ester plasticizer exclusively.

4.4.2 End item examination.

* 4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected on one side only, however, the side shall be alternated on every other roll examined, and visual defects classified as listed below. The same yardage shall be given a through-light inspection for pinholes and lightly coated areas. The through-lighting inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The acceptable quality level (AQL) for this examination, including the pinholes and lightly coated areas found in the through-light examination shall be 4.0 defects per hundred units (yards). The inspection level shall be II. The number of rolls from which the samples are to be selected shall be in accordance with table II. An approximately equal number of yards shall be examined from each roll sampled.

Examine	Defect
Coating	Any cut, hole, tear, scratch, or abrasion mark. Any pinhole. Any uncoated area. Any thinly coated area. Any blister, tunnel, or delamination of coating. Any lump or heavily coated area. Crease or wrinkle that cannot be corrected by manual pressure or resulting in doubling or adhesion of surfaces. Any spot, stain, or streak more than 1 inch in its longest dimension. 1/ Any embedded foreign matter. Any scorch or burn. Any piece not coated on both sides. Face side (lighter coated side) not right hand twill. Color and shade not as specified. Luster not equal to that of the standard sample. Tackiness (coating will adhere to itself, and not readily unroll).
Width	Trimmed width less than minimum specified. Selvages not trimmed. Edges not straight and uniform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

Examine	Defect
General	Objectionable odor. 2/ Not clean. Curled, folded, scalloped, or rolled edges.

1/ Clearly visible at normal inspection distance (approximately 3 feet).

2/ Odors of chemicals commonly used in coating compounds shall not be regarded as objectionable.

TABLE II. Sample size

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample 2/
Up to 1200 inclusive 1/	3	0
1201 to 3200 inclusive	5	0
3201 to 10,000 inclusive	8	0
10,001 to 35,000 inclusive	13	0
35,001 to 150,000 inclusive	20	1
150,001 and over	32	2

1/ If lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination only (see 4.4.2.2).

4.4.2.2 Examination for length.

* 4.4.2.2.1 Examination for length of individual rolls. Each individual roll in the sample shall be examined for the defects listed below. The sample size (number of rolls selected as sample) and the maximum acceptable number of defects shall be as shown in table II. Any of the following shall be considered a defect with respect to length:

Total length of roll less than 150 or more than 175 yards.
Total length on roll is 2 or more yards less than that marked on ticket.
Roll containing more than six pieces.
Any piece in roll less than six yards in length.
Ends of pieces in roll not overlapped, but joined by a seam.

* 4.4.2.2.2 Examination for total yardage in sample. The rolls examined shall be those selected for examination of individual rolls as specified in 4.4.2.2.1. The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total gross lengths marked on the roll tickets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

* 4.4.3 End item testing. The methods of testing contained in FED-STD-191 wherever applicable and as listed in table III shall be followed. The sample unit shall be 3-1/3 continuous yards, full width of the coated cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and up	5

4.4.3.1 Standard test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191, FED-STD-406 or FED-STD-601 will be acceptable except in case of dispute. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

* TABLE III. End item tests

Characteristic	Requirement		Test method
Weight	3.4		5041
Breaking strength	3.4		5100
Tearing strength	3.4	ASTM D	1424
Hydrostatic resistance:			
Initial	3.4		5512 1/
After strength of coating	3.4		5972 2/
After water immersion	3.4		3/
Adhesion of coating:			
Initial	3.4		5970
After water immersion	3.4		4/
Stiffness (warp direction only)	3.4		5204
Blocking	3.4		5872
Resistance to abrasion	3.4.1		5304 5/

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

TABLE III End item tests (cont'd)

Characteristic	Requirement	Test method
Resistance to leakage	3.4.2	5516 6/
Resistance to low temperature	3.4.3	5874 7/
Resistance to accelerated weathering	3.4.4	5804 8/

1/ Water pressure shall be applied to the lightly coated side of the coated cloth only.

2/ Except that the specimen shall be stretched at 20 pounds with the load applied parallel to the twill line only.

3/ The test specimen shall be immersed in water at 70 ± 2° F for 2 hours, then excess water shall be lightly blotted from the surface and the specimen tested immediately thereafter in accordance with Method 5512.

4/ The test specimens used in dry adhesion of coating that have been separated 3 inches, shall be immersed in distilled water at room temperature for 2 hours. After immersion the specimens shall be blotted dry and the wet adhesion shall be determined on the remainder of the specimen in accordance with Method 5970.

5/ The heavily coated side of the cloth (back side) shall be subjected to 10 cycles under a tension of 6 pounds, and under a load of 3 pounds pressure on the test abrader. The abradant shall be "fine" emery cloth. Only three specimens in the warpwise direction shall be tested, and the abrasion resistance shall be determined by visual examination.

6/ Method 5516 shall be used except that the lightly coated side of the cloth (face side) shall contact the water. The hydrostatic head shall be 50 centimeters, and shall be held for a period of 10 minutes. The report shall only include "Measurement of the appearance of water drops".

7/ Method 5874 shall be used, except that the test specified for water resistance shall not be performed, and the test specimens shall be exposed for 35 ± 5 minutes at a temperature of 0° ± 5° F. Testing for requirement specified in 3.4.3 shall take place immediately thereafter without removal from the low-temperature test chamber.

8/ The specimens, with the lightly coated side of the cloth, (face side) facing the light source, shall be exposed in the weatherometer for 200 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

NOTE: Attention is called to contractors who own equipment which operates at one revolution in two hours. This equipment may continue to be used. However, the one revolution per minute shall be the standard procedure in the event of dispute, and the laboratory reports shall state the procedure used.

4.4.4 Examination of preparation for delivery. The examination of preparation for delivery shall be made in accordance with PPP-P-1136 to determine whether packaging, packing amd marking comply with the section 5 requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Coated cloth shall be packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Coated cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth is intended for use in the manufacture of Poncho, Lightweight, with Hood.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Width required (see 3.6).
(c) When laboratory report approval is not required (see 3.1).
(d) First article. When a first article is required (see 3.2.2, 4.3 and 6.4).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard samples, adddress the procuring activity issuing the invitation for bids.

* 6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of five yards of the finished cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

6.5 Coating. Cloth having approximately 1 ounce of coating per square yard on the lightly coated side (face side) have been found to meet the requirements of this specification (see 3.4).

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8305-0807

Review activities:

Army - MD
Navy - MC,
DLA-CT

User activity:

Army - CR

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:03Z
Check the source to verify that this is the current version before use.